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PAPER

Random sampling of self-avoiding theta-graphs

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Abstract

Theta-graphs are a type of spatial graph with two vertices connected by three edges. We investigate embeddings of theta-graphs in the square and simple cubic lattices, using a combination of the Wang-Landau Monte Carlo method with a variant of the BFACF algorithm which accommodates vertices of degree 3. This allows us to estimate the critical exponents governing the number of theta-graphs and the distributions of the different arm-lengths. For the cubic lattice these values can be compared to the corresponding exponents for prime knots. We also study the number of 'monodisperse' theta-graphs where the three arms have the same lengths, and find evidence supporting a conjecture for the critical exponent in two dimensions.

1. Introduction

Self-avoiding walks (SAWs) and polygons (SAPs) on regular lattices are well known models of linear and ring polymers [15, 32, 44]. In three dimensions, in particular, SAWs and SAPs display properties similar to those of real world polymers in a good solvent, such as the critical exponents which govern geometric quantities [15, 32, 44].

In recent years, lattice models of more complicated topologies than linear and ring polymers have received attention. Ring polymers themselves in three dimensions can have non trivial knot type and so the study of the behaviour lattice polygons regarding knot type have been a focus. Another way to generalise the topology of lattice objects (and so the consideration of more complicated polymers) is to allow vertices of degree 3 or more. This can be done in a variety of ways, giving structures like stars [6], watermelons [18, 19], combs [39], tadpoles [25], theta-graphs [40, 49] and dumbbells [26]. See figure 1 for some illustrations.

In this paper, we focus on theta-graphs (*thetas* for short). As graphs, these comprise two vertices and three edges, with each edge connecting one vertex to the other. In three dimensions thetas can have different 'knot' types, which can be grouped according to their crossing number in a similar manner to regular knots, and for which there exist invariants for distinguishing them [48]. While the methods we use here can be applied to thetas of any specified knot type, we focus only on 'unknotted' thetas with crossing number 0 (equivalently, those which can be embedded in a 2-sphere in S^3). This restriction does have some numerical consequences, which we discuss further in section 4.1. See figure 3 for some schematics, and figure 2 for an example of a square lattice theta.

Theta-shaped polymers have been synthesised in laboratories [57], and proteins with a theta topology have been observed [12]. Thetas are one type of a more general class called polymer networks, which characterise important materials including gels and rubbers [23]. Thetas formed by Gaussian chains (rather than lattice SAWs) were studied in [60].

Our initial motivation for this particular study was twofold. Firstly, two-dimensional lattice thetas are one type of object considered by Duplantier [18, 19], who conjectured (among other things) the (entropic) critical exponent γ for polymer networks of fixed topology. In particular, these networks are 'monodisperse', i.e. each 'arm' connecting a pair of vertices has the same length. In this paper we study thetas where the three arms can have varying length ('polydisperse'), but by also considering the monodisperse subset we have been able to check the validity of Duplantier's conjecture (see section 5).

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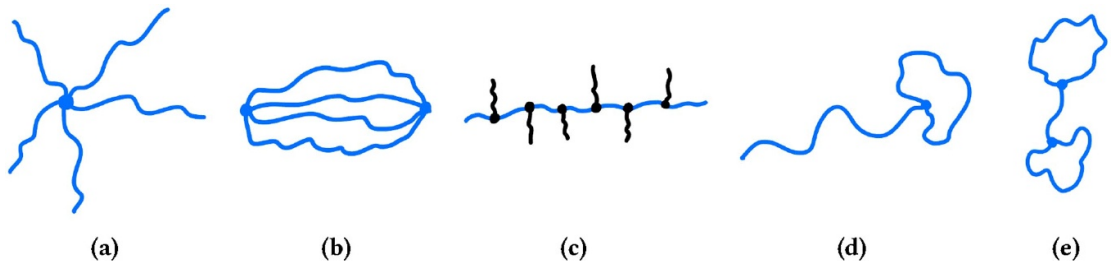


Figure 1. (a) a star, (b) a watermelon, (c) a comb, (d) a tadpole, and (e) a dumbbell.

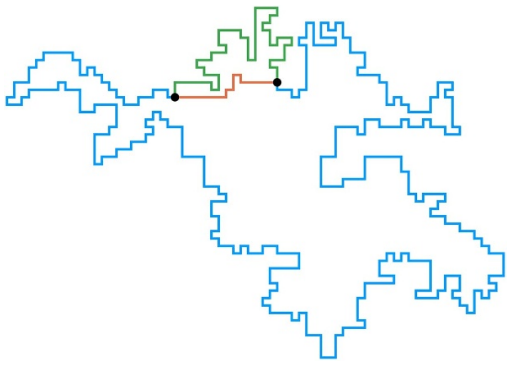


Figure 2. A size 500 theta on the square lattice, with arm lengths 18, 66 and 416.

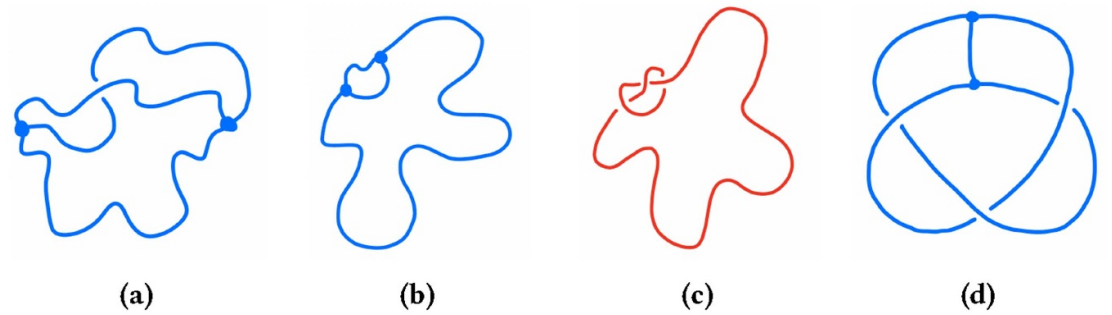


Figure 3. (a) and (b) Two thetas with different distributions of arm lengths. (c) A trefoil knot with a localised knot component. (d) A theta with non-trivial 'knot' type.

Our second motivation is to compare the typical 'shape' of thetas with that of typical knots. Long ring polymers are knotted with high probability [16, 21] and there has been considerable interest in how the topology of polymers like DNA affect biological function (see e.g. the review [58]). One question of interest is whether the 'knotted part' of a typical large prime knot is localised in a small region of the ring, or if it is distributed around the whole structure. Various numerical experiments have shown that knots tend to be 'weakly localised', with the average size of the knotted part of a ring polymer of size n scaling like n^t. Estimates for t have ranged from 0.4 [20], to 0.65 [46, 61] and 0.75 [47, 50]. Numerical experiments with linear chains [59] have also found power-law behaviour, with an exponent around 0.44. We are interested in whether a typical theta looks, in some sense, like a typical knot, with the two shorter arms of a theta comprising a small 'bubble' within a large polygon. See figure 3 for an illustration.

While exact enumeration and series analysis methods for SAWs, SAPs and related objects have been very productive in two dimensions [24], they have generally had limited success in three (or more) dimensions. In these cases Monte Carlo methods have been more effective, and many different algorithms have been developed over the past few decades. These include Beretti-Sokal, the pivot algorithm, PERM, GARM, and GAS. (See the review [31] for descriptions and references for all of these.)

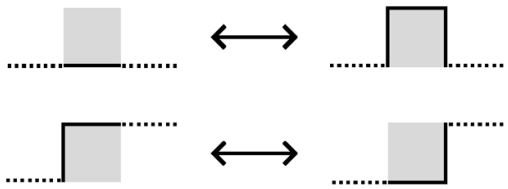


Figure 4. The basic BFACF moves for the square and cubic lattices.

Algorithm 1. (BFACF move for SAPs).

Let P be a SAP defined by the set of edges $E = \{e_1, \ldots, e_n\}$.
procedure BFACF(P)
$e \leftarrow$ uniform sample from E
$q \leftarrow$ uniformly chosen plaquette adjacent to e
$Q \leftarrow$ set of lattice edges adjacent to q
$E' \leftarrow E \sqcup Q$ (i.e. $(E \cup Q) \setminus (E \cap Q)$)
if E' forms the edges of a SAP **then**
$P' \leftarrow E'$
else if E' does not form a SAP **then**
$P' \leftarrow P$
end if
return P'
end procedure

In this paper we combine local BFACF-type moves [5, 13, 14] with the Wang-Landau algorithm [63]. This enables us to sample thetas of fixed knot type (we have focussed on unknots) across a range of shapes and sizes, in order to compute estimates of the number of thetas of a given size, as well as the distribution of the length of shortest or second-shortest arm, or the sum of both. From these data, we calculate estimates for several critical exponents, namely those which govern the number of thetas, the number of monodisperse thetas, the average size of the shortest arm (or second-shortest, or the sum of both), and the mean squared distance between the two branch points.

The structure of the paper is as follows. In section 2 we describe the BFACF algorithm as well as its generalisation to thetas and other branching structures. In section 3 we outline the Wang-Landau algorithm and its use for approximate enumeration. In section 4 we present results for polydisperse theta graphs, including enumerative results and estimates for the distribution of arm-lengths and the separation of the branch points. In section 5 we present results for monodisperse thetas. Finally in section 6 we discuss some future avenues for research.

2. The BFACF algorithm for theta-graphs

The BFACF algorithm [5, 13, 14] is an algorithm for sampling SAPs or SAWs with fixed end-points. It is known to be ergodic for SAPs in the two-dimensional square or triangular lattices; that is, any polygon can be obtained from any other polygon by a sequence of BFACF moves. In the three-dimensional simple cubic, face-centred cubic (FCC) or body-centred cubic lattices, the ergodicity classes for polygons are the knot types [33, 37]. In this work we will focus on the square and cubic lattices; see section 6 for some discussion of other lattices.

The elementary BFACF moves for polygons on the square and cubic are illustrated in figure 4. Note that on the square and cubic lattices, the BFACF moves either leave the total length unchanged, or modify it by ± 2. In algorithm 1 we outline the basic procedure for BFACF moves, as applied to SAPs.

BFACF moves typically form part of the implementation of a Markov chain Monte Carlo study, where the probability of accepting or rejecting a given move is determined by another algorithm. Examples include use with the Metropolis algorithm [38] and with GAS (generalised atmospheric sampling) [35].

Tamaki [55] found a set of BFACF-type moves for spatial graphs (i.e. graphs embedded in space) in the cubic lattice with vertices of degree 2 and 3. In particular, they found that the ergodicity classes are the graph types (generalising the idea of knot types, i.e. two graphs have the same type if they are

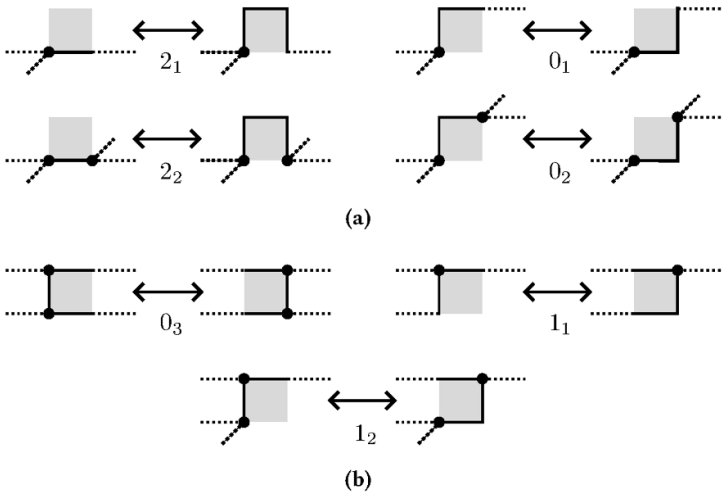


Figure 5. The set of BFACF moves for spatial graphs on the cubic lattice. The moves in (a) do not change the location of the vertices of degree 3, while the moves in (b) do move at least one of the degree 3 vertices. These are in addition to the regular BFACF moves in figure 4(a). This figure is adapted from [55, figure 4.1].

equivalent under ambient isotopy). The set of required moves is illustrated in figure 5. Note that some of these moves change the overall length by ± 1. It is this expanded set of moves which we implement.

While Tamaki did not specifically address the two-dimensional square lattice, we expect the ergodicity classes in the square lattice to also be the graph types.

3. The Wang-Landau method for approximate enumeration

The Wang-Landau method [41, 63] is a Monte Carlo method for estimating the density of states g of a system, defined as

$$g(E) := \frac{\mathrm{d}\Omega(E)}{\mathrm{d}E}, \tag{3.1}$$

where $\Omega(E)$ is the number of states with energy less than or equal to E. Equivalently, the number of states with energy in the small interval $[E, E+\delta E]$ is $N(E) = g(E)\delta E$.

The idea is to perform a random walk in energy space. If s_1 and s_2 are two states with energy levels E_1 and E_2, let $p(s_2|s_1)$ be the probability of proposing a move to s_2 given the current state s_1. Then the probability of accepting such a move has a Metropolis–Hastings form:

$$P(E_1 \to E_2) = \min\left\{\frac{g(E_1)p(s_1|s_2)}{g(E_2)p(s_2|s_1)}, 1\right\}. \tag{3.2}$$

If the move is accepted then we update $g(E_2) \mapsto f \times g(E_2)$ where $f > 1$ is a modification factor; otherwise we update $g(E_1)$ by the same factor. We also maintain a histogram H which tracks the number of visits to each energy level: after the aforementioned accepted/rejected move, we increment either $H(E_1)$ or $H(E_2)$ by 1 as appropriate. In practice the values $g(E)$ become large very quickly, so instead we record $\log g(E)$ and update via $\log g(E) \mapsto \log g(E) + \log f$. A typical initial choice for f is $f_0 = e^1$. We will follow [22] and sometimes refer to the factor $\frac{p(s_1|s_2)}{p(s_2|s_1)}$ as the *Hastings factor* (for many systems, this factor is just 1).

After this algorithm has run sufficiently long, the histogram H should be approximately flat, i.e. all energy levels have been visited approximately equally often. A simple threshold is $\frac{\min(H)}{\max(H)} \geqslant X$ for some fraction X; in this paper we use $X = 90\%$. (One must choose some frequency with which to check for the flatness of H; typically every M moves for some M which depends on the system size.) At this point g will have converged to its true value, within an accuracy proportional to $\log f$. We then decrease f via some function (often $f \mapsto \sqrt{f}$), reset $H = 0$, and then start again.

This process repeats until some criteria have been met: either $f < 1 + \epsilon$ for a small ϵ, or a sufficient (large) number of moves have been made. At this point, $g(E)$ provides an estimate for the *relative* density of states. It must be normalised, typically by knowledge of the actual number $g_{\text{ex}}(E_0)$ of ground states with energy E_0. Then one scales all values by

$$\log g(E) \mapsto \log g(E) - \log g(E_0) + \log g_{\text{ex}}(E_0) \tag{3.3}$$

to obtain the final estimate.

This method works well up to a certain accuracy, but as f continues to decrease a problem can appear. Because we decrease f by $f \mapsto \sqrt{f}$, the value $\log f$ with which we are updating $g(E)$ decreases exponentially. This turns out to be too fast—with $\log f$ decreasing exponentially, the estimates $g(E)$ essentially converge and get 'frozen', with no further improvements to accuracy. See figure 6 for a plot which demonstrates this in the case of SAPs.

A solution to this problem was proposed in [4]. The idea is to decrease $F = \log f$ more slowly: instead of updating $F \mapsto F/2$ when the histogram H is sufficiently flat, we use $F = 1/t$, where t is the Monte Carlo time (proportional to the number of proposed moves). The method given in [4] (and which we use here) is actually two-stage: the exponential updating $F \mapsto F/2$ is used at first, until $F < 1/t$. At that point, we switch to $F = 1/t$, and the histogram H is no longer used. In practice F is still only updated periodically, every M moves for some fixed M. With the error at energy level E defined as [4]

$$\epsilon(E) = \left| 1 - \frac{\log g(E)}{\log g_{\text{ex}}(E)} \right|, \tag{3.4}$$

it is then expected [4] that the average error $\langle \epsilon \rangle$ (across all energy levels) should scale as $\sqrt{F} = t^{-1/2}$.

The Wang-Landau method has seen a wide variety of applications, including analysis of computer networks [1], numerical integration [42], interacting polymers [56, 62] and the Ising model [66]. The method has been adapted in a variety of ways, including parallel implementations [65, 66] and in combination with the N-fold way [45].

3.1. General framework for enumeration

To apply the Wang-Landau method to the approximate enumeration of combinatorial objects like SAPs, we need only make a few small changes. Some of the ideas in this and the following subsections were presented in [22].

First, let $\mathcal{C}$ be a combinatorial class with size function $|\cdot| : \mathcal{C} \to \mathbb{N}_0$. Let $\mathcal{C}_n$ be the set of objects of size n, and define $C_n = |\mathcal{C}_n| < \infty$. (Note that here we use $|\cdot|$ to denote both the size function on $\mathcal{C}$ as well as the usual cardinality of a set.)

Let $N_{\min}$ and $N_{\max}$ be respectively the minimum and maximum sizes of the objects we wish to count. Then the set $\mathcal{S}$ of states is the set of objects with size in the interval $I = [N_{\min}, N_{\max}]$. The 'energy' of an object is its size. Let $B = |\{n \in I : C_n > 0\}|$.

Define a probability distribution ρ on $\mathcal{S}$ where $\rho(\gamma)$ depends only on $|\gamma|$. That is, $\rho(\gamma) = R(|\gamma|)$ where $R : I \to [0, 1]$. Then writing $G(n) = \log g(n)$,

$$\frac{1}{R(n)} = \exp(G(n)) \tag{3.5}$$

now becomes an estimate of the relative multiplicity of $\mathcal{C}_n$. If we know C_m for some $m \in I$, then the multiplicities of the other $\mathcal{C}_n$ can be estimated by

$$\tilde{C}_n = C_m \exp(G(n) - G(m)), \tag{3.6}$$

which is equivalent to (3.3). As with the original Wang-Landau method, we require an irreducible random walk over the objects in $\mathcal{S}$. Then the probability distribution ρ is regularly updated (via the updating of G) according to which states are sampled. The general framework is given in algorithm 2.

Note that instead of setting M to be the total number of samples in algorithm 2, we can instead set a minimum value ϵ for F, replacing the **while** $i < M$ **do** loop with **while** $F > \epsilon$ **do**.

To avoid saturation of errors we can improve this using the $1/t$ algorithm, given in algorithm 3. Again the condition $i < M$ can be replaced by $F > \epsilon$ for some small ϵ.

In figure 6 we compare algorithms 2 and 3 for SAPs on the square lattice. The exact counts p_n are known up to size $N_{\max} = 130$ [11]. We ran the Wang-Landau algorithm on SAPs (combined with BFACF

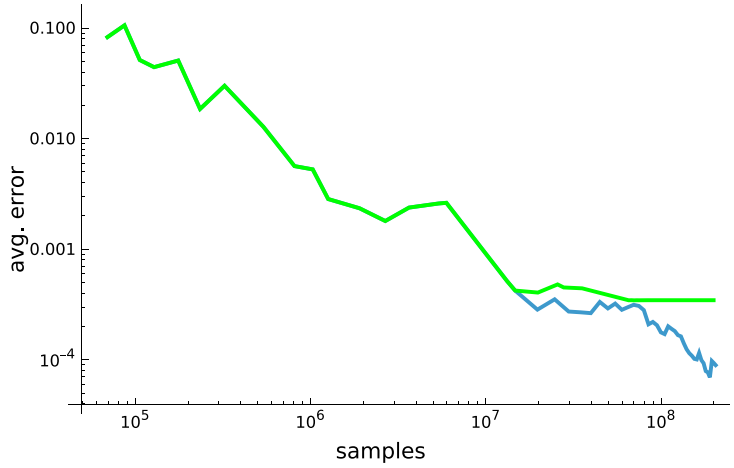


Figure 6. A plot of the average error (3.7) for the Wang-Landau algorithm, sampling SAPs on the square lattice up to size $n_{\max} = 130$, with a total of 2×10^8 samples. Algorithm 2 is in green, while algorithm 3 coincides with algorithm 2 up to $\approx 1.47 \times 10^7$ samples, after which it switches to the '$1/t$ algorithm' and is plotted in blue.

Algorithm 2. (Basic Wang-Landau for enumeration).

Let $\mathcal{C}$ and I be as above. Suppose $m \in I$ is such that C_m is known. Let MetropolisHastings(s, ρ) denote a sample from the distribution ρ given the previous state s. The Wang-Landau algorithm for estimating the quantities C_n is:

```
procedure WangLandau
    s ← initial state
    G ← [0, ..., 0]
    H ← [0, ..., 0]
    F ← 1
    i = 0
    while i < M do                    ▷ M is the total number of samples to take
        s ← MetropolisHastings(s, exp(−G))
        G[s] ← G[s] + F
        H[s] ← H[s] + 1
        i ← i + 1
        if H sufficiently flat then
            H ← [0, ..., 0]
            F ← F/2
        end if
    end while
    C̃ ← C_m exp(G − G(m))
end procedure
```

moves, see section 3.2) up to size $N_{\max}$ until 2×10^8 samples have been taken, periodically computing estimates $\tilde{p}_n$ from G, and computing the average error $\langle \delta \rangle$, where

$$\delta(n) = \left| 1 - \frac{\log \tilde{p}_n}{\log p_n} \right| \tag{3.7}$$

as per (3.4). The saturation of errors in algorithm 2 can clearly be seen in figure 6, where after some time f becomes so small that G (and hence the estimates $\tilde{p}_n$) has essentially converged. After this point there is nothing to be achieved by running the algorithm any longer.

3.2. SAPs

To approximately enumerate (unknotted) SAPs of length n we can use algorithm 2 or algorithm 3. On the square and cubic lattices we have $N_{\min} = 4$. Then

$$C_4 = \begin{cases} 1, & \text{square} \\ 3, & \text{cubic.} \end{cases} \tag{3.8}$$

Algorithm 3. (Improved Wang-Landau for enumeration).

procedure IMPROVEDWANGLANDAU
  $s \leftarrow$ initial state
  $G \leftarrow [0, \ldots, 0]$
  $H \leftarrow [0, \ldots, 0]$
  $F \leftarrow 1$
  $i \leftarrow 0$
  stage $\leftarrow 1$
  while $i < M$ **do**
    $s \leftarrow$ METROPOLISHASTINGS$(s, \exp(-G))$
    $G[s] \leftarrow G[s] + F$
    $H[s] \leftarrow H[s] + 1$
    $i \leftarrow i + 1$
    if *stage* $= 1$ and H sufficiently flat **then**
      $H \leftarrow [0, \ldots, 0]$
      $F \leftarrow F/2$
      if $F < B/i$ **then**
        $F \leftarrow B/i$
        stage $\leftarrow 2$
      end if
    else if *stage* $= 2$ **then**
      $F \leftarrow B/i$
    end if
  end while
  $\tilde{C} \leftarrow C_m \exp(G - G(m))$
end procedure

It remains to define the random walk over the set of SAPs of size $\leq N_{\max}$. This is where the BFACF algorithm comes in (see algorithm 1).

Note that when implementing BFACF moves in Wang-Landau, care must be taken with the length-changing moves: there is only one way to choose an edge and adjacent plaquette when increasing the length, but there are multiple ways to choose an edge/plaquette when decreasing the length. When using Wang-Landau this difference can be accounted for in the Hastings factor. For the square and cubic lattices, we have (when $|S| = n$)

$$\frac{p(S|S')}{p(S'|S)} = \begin{cases} \frac{3n}{n+2} & \text{if increasing length by 2} \\ \frac{n}{3(n-2)} & \text{if decreasing length by 2} \\ 1 & \text{if length is unchanged.} \end{cases} \tag{3.9}$$

An alternative strategy is to restrict the types of length-decreasing moves, so that they are exactly in one-to-one correspondence with length-increasing moves. This simplifies the Hastings factors slightly (it eliminates the factors of 3 and 2 in the numerators and denominators) but does mean that certain choices of e and q in algorithm 1 will be immediately rejected.

3.3. Spatial graphs and multi-statistic Wang-Landau

Algorithm 1 can be generalised to spatial graphs with vertices of degree 3 in a straightforward manner. One still chooses an edge e and an adjacent plaquette q uniformly at random.

- If q is not incident on any vertex of degree 3, then proceed as in algorithm 1. The Hastings factor is the same as for SAPs.
- If q is incident on one or two vertices of degree 3, then one of the moves illustrated in figure 5 will be proposed. The moves in figure 5(a) do not shift a vertex of degree 3 and thus the updating is the same as in algorithm 1 (again the Hastings factor is the same as for SAPs). The moves in figure 5(b) do shift one or more vertices of degree 3, and hence the updating is not as simple as 'inverting' the edges around the plaquette. The Hastings factor is still straightforward to compute.
- If q is incident on three or more vertices of degree 3 (this is of course not possible with thetas), no move is proposed.

For the enumeration of thetas it then suffices to take this generalisation of algorithm 1 and use it within either algorithm 2 or algorithm 3. However here we are interested not only in counting thetas, but also understanding the behaviour of the different arm-lengths. For this purpose we thus need something more general than algorithms 2 and 3. To do so, we make the following modifications to the framework of section 3.1:

- We generalise the size function on the combinatorial class $\mathcal{C}$ to a function $m : \mathcal{C} \to (\mathbb{N}_0)^2$, where $m(\gamma) = (|\gamma|, s(\gamma))$ and $s(\gamma)$ is some statistic we wish to measure (for our purposes it will be the number of edges in one or a pair of the arms of γ). Then let $\mathcal{C}_{n,k} = \{\gamma \in \mathcal{C} : m(\gamma) = (n,k)\}$ and $C_{n,k} = |\mathcal{C}_{n,k}|$.
- We then also generalise the set I and the number B, using

$$I = \{(n,k) \in (\mathbb{N}_0)^2 : N_{\min} \leq n \leq N_{\max}, C_{n,k} > 0\}, \qquad B = |I|. \tag{3.10}$$

  Note that thetas are sufficiently simple that we can explicitly compute I and B for all cases considered in this paper. It is also straightforward to compute $C_{N_{\min},k}$.
- The probability distribution ρ on $\mathcal{S}$ is now such that $\rho(\gamma)$ depends on $m(\gamma)$, and with $\rho(\gamma) = R(n,k)$, we have

$$\frac{1}{R(n,k)} = \exp(G(n,k)) \tag{3.11}$$

  being an estimate of the relative multiplicity of $\mathcal{C}_{n,k}$. Here $G(n,k) = \log g(n,k)$, where g is still the density of states, which now are indexed by two integers (n,k).
- The Metropolis–Hastings probabilities are modified in the obvious way: if s_1 and s_2 are two states with $m(s_1) = (n_1,k_1)$ and $m(s_2) = (n_2,k_2)$, then

$$P((n_1,k_1) \to (n_2,k_2)) = \min\left\{\frac{g(n_1,k_1)p(s_1|s_2)}{g(n_2,k_2)p(s_2|s_1)}, 1\right\}. \tag{3.12}$$

- The arrays G and H are now indexed by n and k instead of just n.

The algorithm is then run in the same way, with the goal being a flat histogram across all reachable values of $(n,k) \in I$. This works within algorithms 2 and 3.

We note here that while the ergodicity of the BFACF algorithm for SAPs is well understood, it does not necessarily follow from the results of Tamaki [55] that every possible theta γ (of a given fixed topology) with $m(\gamma) \in I$ is reachable *when n is restricted to* $n \leq N_{\max}$. That is, it may be the case that some γ can only be reached by first sampling some γ' with $|\gamma'| > N_{\max}$, and then shrinking. Here we will simply be assuming that this is not the case, or if it is, then the number of such thetas is negligible.

4. Results: polydisperse theta-graphs

For our purposes a *theta-graph* (*theta* for short) is a spatial graph with two vertices of degree 3, which we denote v_1 and v_2, arranged so that there are three 'arms' connecting v_1 and v_2. In three dimensions there are actually infinitely many topologically distinct spatial graphs which satisfy this property, just as there are infinitely many knot types for a simple closed curve. See for example [48] for further work on knot types of theta graphs, and see figure 3 for some illustrations.

Let $\mathcal{T}_n^{(d)}$ be the set of theta-graphs with n edges in the d-dimensional hypercubic lattice (counted up to translation), and let $t_n^{(d)} = |\mathcal{T}_n^{(d)}|$. Similarly let $\Theta_n^{(d)}$ be the set of unknotted thetas of size n, and $\theta_n^{(d)} = |\Theta_n^{(d)}|$. Note that $\Theta_n^{(2)} = \mathcal{T}_n^{(2)}$ for all n. We will generally omit the (d) superscript when it is clear that we are working on the square or cubic lattices.

Because the BFACF algorithm preserves spatial graph type, in this work we will generally restrict to unknotted thetas, as per figure 3. However, we do note that different types of thetas can also be sampled using the BFACF algorithm. We expect that thetas with other knot types will have different critical exponents, and we are not sure what other kind of quantitative differences one might expect to find. For the remainder of the paper we will usually just use 'theta' to refer to unknotted theta-graphs, unless otherwise specified.

On a given lattice we say that a theta is *polydisperse* if the three arms connecting v_1 and v_2 may have different lengths. If instead we restrict to thetas where all three arms have the same length, we say such objects are *monodisperse* or *uniform* (see section 5).

For each of the square and cubic lattices we ran the Wang-Landau algorithm in three different ways:

(I) Flattening across the size n and the length ℓ_1 of the shortest arm;

(II) Flattening across the size n and the length ℓ_2 of the second-shortest (equivalently, second-longest) arm;

(III) Flattening across the size n and the sum $\ell_1 + \ell_2$ of the shortest and second-shortest arms.

For (I)–(III) we sampled thetas up to size 500. For each lattice and for each of (I)–(III) we ran 20 independent threads. In each thread we collected samples after every 10 attempted BFACF moves, up to a total of 2×10^{10} samples.

4.1. Enumeration

Our first goal is to estimate the number $\theta_n^{(d)}$ of thetas of size n on each of the lattices in question. This can be achieved using any of (I)–(III) above, summing over ℓ_1, ℓ_2 or $\ell_1 + \ell_2$ respectively. Each thus gives a different estimate of θ_n.

The known terms of the sequences $(\theta_n^{(d)})$ are (to the best of our knowledge)

$$(\theta_n^{(2)})_{n\geqslant 7} = (2, 0, 12, 6, 62, 60, 338, 430, 1966, 2794, 11772, 17898, 71390, 114496, 438112, 731698, 2718114, 4681116, 17013354, 30025926, 107283688, 193174670, \dots) \tag{4.1}$$

for the square lattice, and

$$(\theta_n^{(3)})_{n\geqslant 7} = (18, 24, 344, 582, 5934, 12120, 104250, 239610, 1877626, 4655982, \dots) \tag{4.2}$$

for the cubic lattice. These were computed using a basic backtracking algorithm. Note that $\theta_n^{(2)} = t_n^{(2)}$ for all n; we expect that the numbers quoted in (4.2) (up to size $n = 16$) are the same for $t_n^{(3)}$ and $\theta_n^{(3)}$, since the smallest knotted *polygon* on the cubic lattice has size 24 [17].

Recall that with p_n denoting the number of SAPs (of any knot type) of perimeter n on a given lattice, it is widely expected that

$$p_n = C\mu^n n^{\alpha-3}(1 + o(1)). \tag{4.3}$$

Unknots are known [53] to have a strictly smaller growth rate $\mu_0 < \mu$, and numerical evidence [34] suggests that the exponent α_0 is the same as for all polygons. For other fixed knot types K, it is conjectured [2, 34] that $\mu_K = \mu_0$ and $\alpha_K = \alpha_0 + f_K$, where f_K is the number of prime knot components of K. For knots in very narrow tubes of the cubic lattice, this result has been proved [3].

In two dimensions it is believed that the critical exponent $\alpha = \frac{1}{2}$, while in three dimensions $\alpha \approx 0.23721$ [24]. (There is no reason to believe the latter value is rational, or even algebraic.) The value μ is referred to as the *connective constant* (or sometimes *growth constant*). For the square and cubic lattices, the current best estimates are

$$\mu \approx \begin{cases} 2.63815853032790(3), & \text{square [29]} \\ 4.684039931(27), & \text{cubic [9]} \end{cases} \tag{4.4}$$

where the values in brackets indicate uncertainty in the final digit.

The lower order terms are expected to have correction-to-scaling exponents:

$$p_n = C\mu^n n^{\alpha-3}\left(1 + \frac{a_1}{n} + \frac{a_2}{n^2} + \dots + \frac{b_0}{n^{\Delta_1}} + \frac{b_1}{n^{\Delta_1+1}} + \dots\right) \tag{4.5}$$

for a universal exponent Δ_1. Numerical evidence [7, 24] suggests that $\Delta_1 = \frac{3}{2}$ in two dimensions. We are not aware of numerical estimates for Δ_1 using three dimensional self-avoiding *polygons*, but work on self-avoiding *walks* [10] has estimated $\Delta_1 \approx 0.528(8)$.

For thetas it is known [26, 52] that t_n has the same exponential growth rate μ, that is,

$$\lim_{n\to\infty} \frac{1}{n} \log t_n = \log \mu. \tag{4.6}$$

For unknotted thetas on the cubic lattice, we have the following result. (We are not aware of this exact result appearing anywhere in the literature.)

Lemma 1. *Unknotted thetas on the cubic lattice have the same growth rate as unknotted polygons, that is,*

$$\lim_{n\to\infty}\frac{1}{n}\log\theta_n^{(3)} = \log\mu_0. \tag{4.7}$$

Proof. Let $p_n(0_1)$ be the number of unknotted polygons with n edges (n must be even). For the lower bound, we show that any unknotted polygon of size n can be converted into an unknotted theta of size $n+3$, and another of size $n+6$. To do this, let P be a polygon of size n, and let v be the lexicographically largest vertex of P (i.e. first find all vertices with maximum x-coordinate, then among those maximise the y-coordinate, then the z-coordinate). At least one of the vertices $v-(0,1,0)$ and $v-(0,0,1)$ must be adjacent to v in P; for now, assume that $w = v-(0,0,1)$ is one of them.

Now define the two theta-graphs P' and P'' as follows:

- P': Take all the edges of P, as well as the path with steps $+x, +z, -x$ from w to v. Then $|P'| = n+3$, and the two vertices of degree 3 in P' are v and w.
- P'': Let $u = w+(1,0,0)$. Take all the edges of P except the one between v and w. Add the edge from w to u, and then the path $+z, -x$ from u to v, and the path $+y, +z, -x, -y$ from u to v. Then $|P''| = n+6$, and the two vertices of degree 3 in P'' are v and u.

This process is clearly reversible, and gives for $n\geqslant 10$

$$\theta_n^{(3)} \geqslant p_{n-q}(0_1), \qquad \text{where } q = \begin{cases} 3 & n \text{ odd}, \\ 6 & n \text{ even}. \end{cases} \tag{4.8}$$

(If w was not adjacent to v in P, then the vertex $w' = v-(0,1,0)$ can be used instead, with a similar set of edges added.)

Now it follows from (4.8) that

$$\liminf_{n\to\infty}\frac{1}{n}\log\theta_n^{(3)} \geqslant \liminf_{n\to\infty}\frac{1}{n}\log p_{n-3}(0_1) = \log\mu_0 \tag{4.9}$$

where the limit is taken through odd values of n. Similarly,

$$\liminf_{n\to\infty}\frac{1}{n}\log\theta_n^{(3)} \geqslant \liminf_{n\to\infty}\frac{1}{n}\log p_{n-6}(0_1) = \log\mu_0 \tag{4.10}$$

where the limit is taken through even values of n. These two limits establish the lower bound.

For the upper bound, we appeal to [43, Corollary 2.3]. In that result, $h^0(n,\tau)$ is the number of embeddings of the graph τ in $\mathbb{Z}^3$, where every cycle of τ is unknotted. By taking τ to be theta-graphs, we have $\theta_n^{(3)} \leq h^0(n,\tau)$, and then by [43, Corollary 2.3]

$$\limsup_{n\to\infty}\frac{1}{n}\log\theta_n^{(3)} \leq \limsup_{n\to\infty}\frac{1}{n}\log h^0(n,\tau) = \log\mu_0. \tag{4.11}$$

The result (4.7) follows. □

As mentioned above, in this work we are considering unknotted thetas in the cubic lattice. As a result of lemma 1 it is more appropriate to use the growth rate μ_0 of unknots instead of μ. Numerical estimates indicate that $\log\mu - \log\mu_0 \approx 4.15\times10^{-6}$ [30, 36][1]. We will generally be using μ_0 for the remainder of the paper (of course in 2D $\mu = \mu_0$).

We will assume that θ_n has a similar subexponential factor form to polygons:

$$\theta_n \sim B\mu_0^n n^{\zeta}. \tag{4.12}$$

for constants B and ζ. (In this work we have made no attempt to estimate B.)

There is little work in the literature on the number of lattice thetas. They appear in Sykes' 'counting theorem' [54], which relates the numbers of SAWs, polygons, thetas, tadpoles and figure-eights.

[1] Though for the lengths of thetas that we are working with, this difference has essentially no effect on any of our calculations.

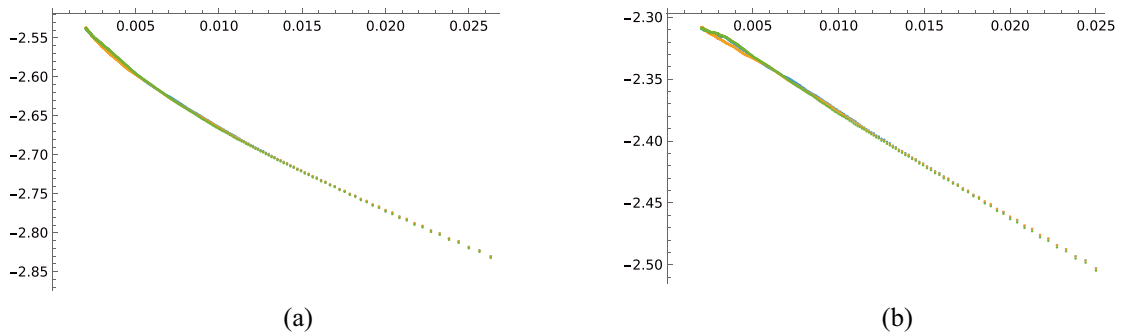


Figure 7. (a) A plot of $L_n^* - n\log\mu_0 - (\alpha-2)\log n$ for the square lattice against $\frac{1}{n}$. The data are from (I) (blue), (II) (orange) and (III) (green), taken by first averaging over the 20 independent Wang-Landau runs. (b) The same data, except with the value -1.459 used instead of $(\alpha-2)$.

Some numerical work was done in [26] on the square and triangular lattices, resulting in the estimate $\zeta = -1.35 \pm 0.15$ (they use the symbol δ, corresponding to $\zeta = \delta - 1$). Enumeration of thetas on the hexagonal lattice was used in [28] but asymptotics were not computed at the time. Some short series have also been provided to us [27].

As will be seen in section 4.2, a typical theta-graph of size n tends to have two short arms of size $o(n)$ and one long arm of size $\approx n$. Roughly speaking, it follows that a typical theta resembles a SAP of size $\approx n$ with a small loop of size $o(n)$ 'inserted' somewhere. Hence one might expect that $\theta_n \approx cnp_n$ for constant $c = \frac{B}{C}$. Combining (4.3) and (4.12) would then give $\zeta = \alpha - 2$, i.e. $\zeta = -\frac{3}{2}$ in 2D and $\zeta \approx -1.76279$ in 3D. We note that the value $-\frac{3}{2}$ is just within the range suggested in [26].

To analyse the data we will assume the basic asymptotic form (4.12) holds, or potentially one with the correction-to-scaling factor

$$\theta_n \sim B\mu_0^n n^\zeta \left(1 + \frac{a}{n^\Delta}\right) \tag{4.13}$$

for a constant a, where we take $\Delta = 1$ for 2D and $\Delta = \frac{1}{2}$ for 3D. In fact, because the Wang-Landau method really produces estimates for $\log\theta_n$, we will actually use the logs of (4.12) and (4.13)

$$L_n = \log\theta_n \sim \log B + n\log\mu_0 + \zeta\log n + \frac{a}{n^\Delta} \tag{4.14}$$

where we have used $\log(1 + \frac{a}{n^\Delta}) \sim \frac{a}{n^\Delta}$.

Square lattice

For the square and cubic lattices we essentially have two sequences—one for even n and one for odd n. We expect μ_0 and ζ to be the same for both, but B and a may depend on the parity of n. We can analyse these separately, but it is also fruitful to analyse the median sequence

$$L_n^* := \frac{1}{2}L_n + \frac{1}{2}L_{n+1} \tag{4.15}$$

$$\sim \log B^* + n\log\mu_0 + \zeta\log n + \frac{a^*}{n^\Delta} \tag{4.16}$$

where B^* and a^* are constants which depend on $B_{\text{even}}, B_{\text{odd}}, a_{\text{even}}, a_{\text{odd}}$, and μ_0.

We first test the proposition that $\zeta_{\text{sq}} = \alpha_{\text{sq}} - 2 = -\frac{3}{2}$ by plotting $L_n^* - n\log\mu_0 - (\alpha-2)\log n$ against $\frac{1}{n}$. If indeed $\zeta = \alpha - 2$ then (assuming $\Delta = 1$) this plot should look linear. In figure 7(a) this quantity is plotted (using data from (I)–(III), averaged over the independent Wang-Landau runs) and it is clearly not linear in $\frac{1}{n}$. On the other hand, by testing different values of ζ and minimising the sum of the residuals between the data and a linear fit, we find $\zeta_{\text{sq}} \approx -1.459$ results in a quite straight plot (see figure 7(b)). Due to the somewhat ad-hoc nature of this particular method, we have not attempted to produce an error bar for this estimate of ζ.

We also tried directly fitting curves of the form $\log B^* + \zeta\log n$ and $\log B^* + \zeta\log n + \frac{a^*}{n}$ to $L_n^* - n\log\mu_0$, with mixed results. We separately took the data generated by (I)–(III) and fitted curves to these using MATHEMATICA"s LINEARMODELFIT function (with the default 95% confidence intervals) for values of n in $[n_{\min}, 499]$. We varied $n_{\min}$ over the range $[10, 300]$ for the basic asymptotic form and $[10, 150]$ using the correction-to-scaling term. See figure 8(a) for plots of the estimated values of ζ, plotted against $\frac{1}{n_{\min}}$. Unfortunately these fits have not yielded particularly precise estimates.

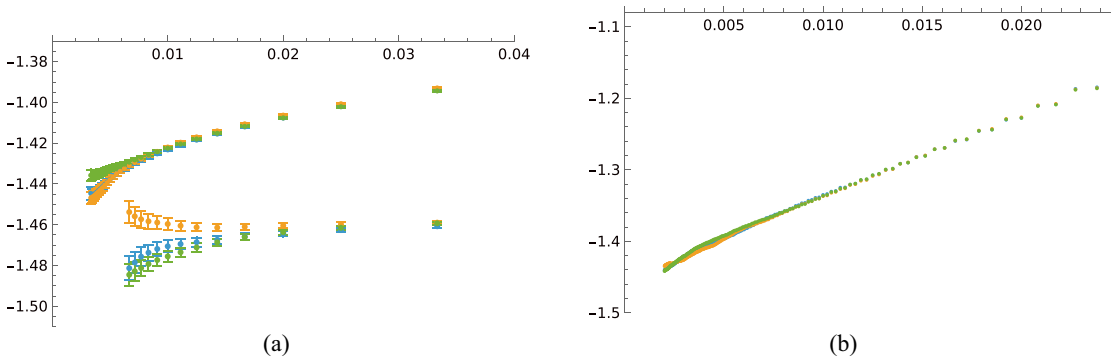


Figure 8. (a) Plots of the estimated values of ζ for the square lattice, using median data from (I) (blue), (II) (orange) and (III) (green). These are obtained using MATHEMATICA"s LINEARMODELFIT function, fitting $L_n^* - n\log\mu_0$. The top three sets are fit without any correction-to-scaling term, while the bottom three do include the $\frac{a}{n^\Delta}$ term, with $\Delta = 1$. The horizontal axis is $\frac{1}{n_{\min}}$. (b) A plot of R_n^* as per (4.17), for $n \in [20, 499]$, using data from (I) (blue), (II) (orange) and (III) (green) computed by first averaging over the 20 independent Wang-Landau runs. The horizontal axis is $\frac{1}{n}$.

Table 1. Estimates for ζ for square lattice thetas, using fits of R_n^* against $\frac{1}{n}$ in various windows.

Fit range	Estimate for ζ
$n \in [50, 499]$	-1.4582 ± 0.0004
$n \in [100, 499]$	-1.4620 ± 0.0006
$n \in [150, 499]$	-1.4653 ± 0.0009
$n \in [200, 499]$	-1.4677 ± 0.0013
$n \in [250, 499]$	-1.4710 ± 0.0020
$n \in [300, 499]$	-1.4717 ± 0.0032
$n \in [350, 499]$	-1.471 ± 0.005
$n \in [400, 499]$	-1.470 ± 0.011
fit range	estimate for ζ
$n \in [50, 150]$	-1.4489 ± 0.0010
$n \in [100, 200]$	-1.4528 ± 0.0017
$n \in [150, 250]$	-1.4591 ± 0.0025
$n \in [200, 300]$	-1.457 ± 0.004
$n \in [250, 350]$	-1.469 ± 0.006
$n \in [300, 400]$	-1.473 ± 0.008
$n \in [350, 450]$	-1.470 ± 0.009
$n \in [400, 499]$	-1.470 ± 0.011

Another method for estimating ζ is to note that

$$R_n^* := \frac{1}{\log 2}\left(L_n^* - L_{n/2}^* - \frac{n}{2}\log\mu_0\right) \sim \zeta + \frac{a^\dagger}{n^\Delta} \tag{4.17}$$

for a constant $a^\dagger$ which depends on a^* and Δ. (Note that we need to use L_n^* here instead of L_n, otherwise the difference between B_{even} and B_{odd} introduces a further term.) In figure 8(b) we plot R_n^* for $n \in [20, 499]$ using data from (I)–(III), computed by first averaging over the 20 independent Wang-Landau runs. The three sets of data are quite linear in $\frac{1}{n}$, as expected.

To quantify this, we used LINEARMODELFIT to fit R_n^* against $\frac{1}{n}$ as plotted in figure 8(b), for n in the windows $[n_{\min}, 499]$ as well as $[n_{\min}, n_{\min} + 100]$, where we used $n_{\min} = 50, 100, \ldots, 400$. The resulting estimates for ζ, along with the confidence intervals, are presented in table 1.

It should be noted that the estimates of θ_n, etc, are more accurate for small n, but that asymptotic behaviour may not become evident until n is quite large[2]. Based on these fits, we think the estimate $\zeta_{\text{sq}} = -1.470 \pm 0.011$ to be reasonable, as it encompasses most of the error bars in table 1.

[2] An alternative fitting method (which produces virtually identical estimates) is to take a linear fit of nR_n^* against n and estimate the slope.

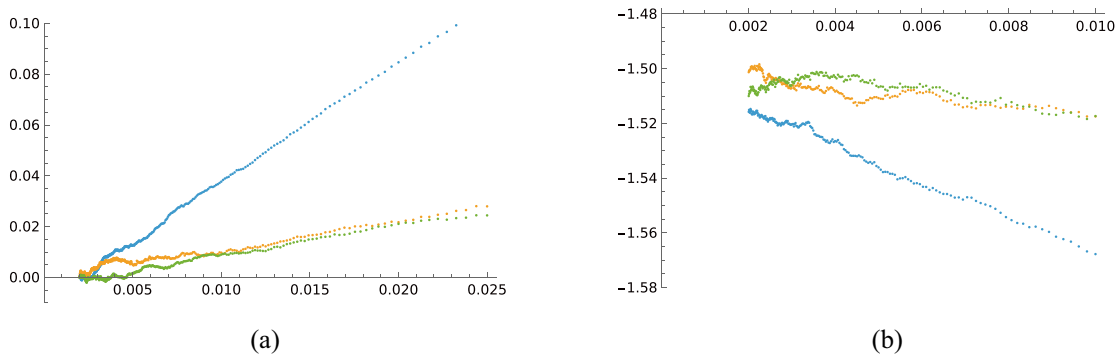


Figure 9. (a) Plots of $\underline{\theta}_n^{[1]*} - n\log\mu_0 - (\alpha-2)\log n$ (blue) and similarly for $\underline{\theta}_n^{[2]*}$ (orange) and $\underline{\theta}_n^{[12]*}$ (green), all against $\frac{1}{n}$, for the square lattice. These have been shifted vertically so that the final term is 0 (so that they all fit on the same plot). The plots are linear as expected. (b) A similar plot to figure 8(b), using ratios to estimate λ. (The colour scheme is the same as (a).) Using linear fits over $n \in [100, 499]$, the projected intercepts are -1.501 (blue), -1.502 (orange) and -1.499 (green).

It seems quite likely to us that $\zeta_{\text{sq}} > -\frac{3}{2}$, that is, $\zeta_{\text{sq}} > \alpha - 2$ and our rough calculation earlier in this section was missing some details. It is, of course, possible that with samples of much larger thetas, estimates of ζ may yet get closer to $\alpha - 2$. We also note that virtually every exponent related to 2D SAPs, SAWs, etc., is a rational number whose denominator is a power of 2; however, at this point we do not have a sufficiently precise estimate for ζ_{sq} to conjecture such a value.

To shed some further light on the possibility that $\zeta_{\text{sq}} > \alpha - 2$, we can attempt to investigate the behaviour of the number of thetas whose shortest arm (or second-shortest, or both) is as small as possible. These may be more likely to look like a large SAP with a very small loop inserted somewhere.

For a given theta-graph T, let $(\ell_1(T), \ell_2(T), \ell_3(T))$ be the number of edges in the three arms, with $\ell_1 \leq \ell_2 \leq \ell_3$. Then, we define the counting sequences for thetas with ℓ_1, ℓ_2 or $\ell_1 + \ell_2$ minimal:

$$\underline{\theta}_n^{[1]} = |\{T \in \mathcal{T}_n : \ell_1(T) = \tfrac{1}{2}(3 + (-1)^n)\}| \tag{4.18}$$

$$\underline{\theta}_n^{[2]} = |\{T \in \mathcal{T}_n : \ell_2(T) = \tfrac{1}{2}(5 - (-1)^n)\}| \tag{4.19}$$

$$\underline{\theta}_n^{[12]} = |\{T \in \mathcal{T}_n : \ell_1(T) + \ell_2(T) = 4\}|. \tag{4.20}$$

(These definitions work for both the square and cubic lattices.) Then we expect

$$\underline{\theta}_n^{[1]} \sim C\mu_0^n n^{\lambda} \tag{4.21}$$

for an exponent λ, which is likely to be $\alpha - 2$. Similar behaviour should hold for $\underline{\theta}_n^{[2]}$ and $\underline{\theta}_n^{[12]}$. As before we also use the $*$ superscript to denote the median sequences.

In figure 9(a) we plot $\underline{\theta}_n^{[1]*} - n\log\mu_0 - (\alpha-2)$ (blue) and similarly for $\underline{\theta}_n^{[2]*}$ (orange) and $\underline{\theta}_n^{[12]*}$ (green). As expected, the data are quite linear in $\frac{1}{n}$, suggesting that $\alpha - 2$ is the correct exponent. In figure 9(b) we plot the analogous quantity to (4.17). While the plots are not as clean as those in figure 8(b), we think it very likely that $\lambda = \alpha - 2 = -\frac{3}{2}$. So this contrasts with our estimate of ζ_{sq}, further suggesting that the argument that ζ is $\alpha - 2$ is an oversimplification.

Cubic lattice

For the cubic lattice we repeat the calculations from above. We initially work with the assumption that $\Delta \approx \frac{1}{2}$. In figure 10 we plot similar quantities to figures 7(a) and 8(b), plotting against $\frac{1}{\sqrt{n}}$. The plot in figure 10(a) displays some curvature; changing the exponent from $\alpha - 2$ to something larger did not yield a nicely straight curve.

Figure 10(b) tells a clear picture: R_n^* cannot be approximated by a linear function in $\frac{1}{n^{1/2}}$, at least not for $n \leq 500$. We thus introduce a second correction-to-scaling term, and assume

$$R_n^* \approx \zeta + \frac{a^\dagger}{n^{1/2}} + \frac{b^\dagger}{n} \tag{4.22}$$

where it may be the case that $|a^\dagger|$ is quite a bit smaller than $|b^\dagger|$[3]. We summarise the results of a few such fits in table 2.

[3] It is of course also possible that θ_n exhibits entirely different asymptotic behaviour, including a possible logarithmic term. We have investigated other kinds of fits, including logarithmic factors, with little success. For $n \leq 500$ and with coarse Monte Carlo data, it is very difficult to distinguish between $\log n$ and small powers of n.

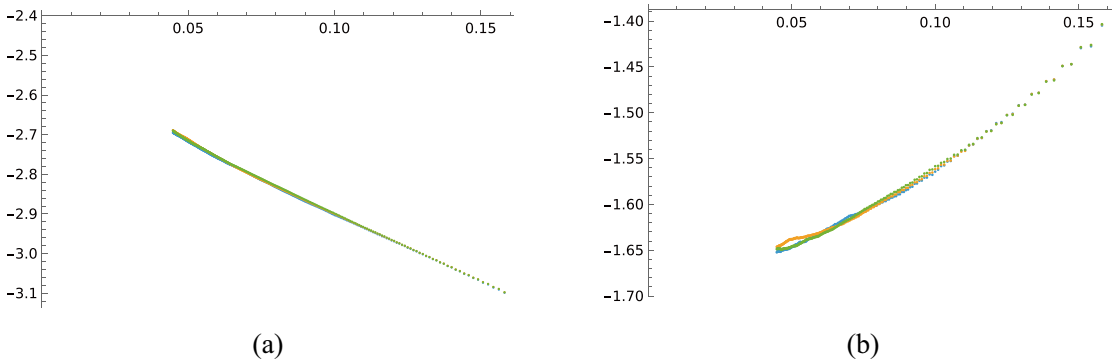


Figure 10. (a) A plot of $L_n^* - n\log\mu_0 - (\alpha-2)\log n$ for the cubic lattice against $\frac{1}{\sqrt{n}}$. The data are from (I) (blue), (II) (orange) and (III) (green), taken by first averaging over the 20 independent Wang-Landau runs. (b) A plot of R_n^* as per (4.17), using data from (I) (blue), (II) (orange) and (III) (green) computed by first averaging over the 20 independent Wang-Landau runs. The horizontal axis is $\frac{1}{\sqrt{n}}$.

Table 2. Estimates for ζ, $a^\dagger$ and $b^\dagger$ for the cubic lattice, as per (4.22).

Fit range	Estimate for ζ	Estimate for $a^\dagger$	Estimate for $b^\dagger$
$n \in [50, 499]$	-1.6824 ± 0.0020	0.34 ± 0.05	8.65 ± 0.31
$n \in [100, 499]$	-1.679 ± 0.004	0.25 ± 0.13	9.4 ± 0.9
$n \in [150, 499]$	-1.667 ± 0.008	-0.19 ± 0.27	13.0 ± 2.2
$n \in [200, 499]$	-1.657 ± 0.016	-0.5 ± 0.6	16 ± 5

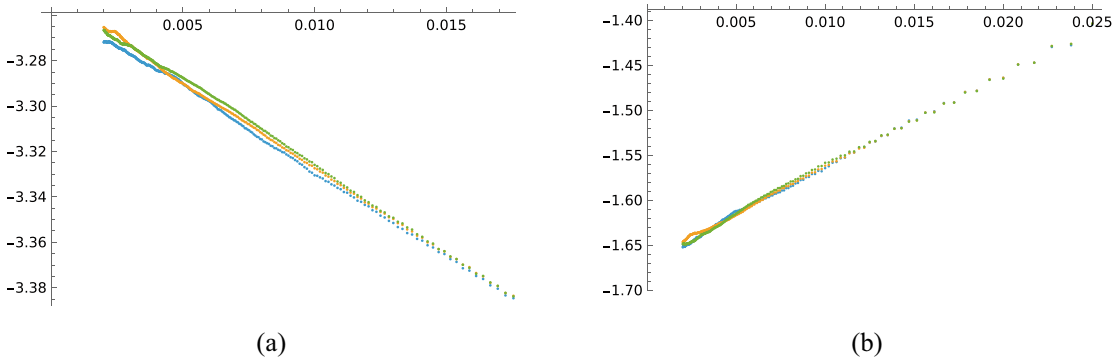


Figure 11. (a) A plot of $L_n^* - n\log\mu_0 - (-1.67)\log n$ for the cubic lattice against $\frac{1}{n}$. The data are from (I) (blue), (II) (orange) and (III) (green), taken by first averaging over the 20 independent Wang-Landau runs. (b) A plot of R_n^* as per (4.17), using data from (I) (blue), (II) (orange) and (III) (green) computed by first averaging over the 20 independent Wang-Landau runs. The horizontal axis is $\frac{1}{n}$.

The data in table 2 suggest that it is possible that $a^\dagger = 0$, that is, $R_n^* \sim \zeta + \frac{b^\dagger}{n}$. To further investigate this, in figure 11 we have plotted the data against $\frac{1}{n}$ instead of $\frac{1}{\sqrt{n}}$. For (a) we tried different values of ζ in order to minimise the sum of the residuals between the data and a linear fit, and settled on -1.670 as the best value. In (b) we plot R_n^* against $\frac{1}{n}$, and found a reasonably straight plot. See table 3 for the results of fits against $\frac{1}{n}$.

Based on the above plots and estimates we are unable to determine whether $a^\dagger$ is small but nonzero, or if in fact $a^\dagger = 0$ and the largest correction-to-scaling term has the form $\frac{1}{n}$[4]. In either case there is substantial overlap in the estimates for ζ_{cu}, and based on the above we estimate $\zeta_{\text{cu}} = -1.67 \pm 0.015$. This is quite different to $\alpha - 2 \approx -1.763$.

We also looked at the behaviour of $\underline{\theta}_n^{[1]*}$, $\underline{\theta}_n^{[2]*}$ and $\underline{\theta}_n^{[12]*}$ for the cubic lattice. We have omitted the plots for brevity. We again found that plotting against $\frac{1}{n}$ yielded straighter plots than $\frac{1}{\sqrt{n}}$. Whether or not the largest correction-to-scaling term is $\frac{1}{n}$ or $\frac{1}{\sqrt{n}}$, the exponent λ does appear to be close to $\alpha - 2$.

[4] We should remind the reader that we are studying *unknotted* thetas here. Unknotted polygons have a different critical point to all polygons, and we are not aware of any predictions for correction-to-scaling terms specifically for unknotted polygons.

Table 3. Estimates for ζ for cubic lattice thetas, using fits of R_n^* against $\frac{1}{n}$ in various windows.

Fit range	Estimate for ζ
$n \in [50, 499]$	-1.6712 ± 0.0005
$n \in [100, 499]$	-1.6718 ± 0.0006
$n \in [150, 499]$	-1.6721 ± 0.0008
$n \in [200, 499]$	-1.6719 ± 0.0011
$n \in [250, 499]$	-1.6713 ± 0.0016
$n \in [300, 499]$	-1.6715 ± 0.0025
$n \in [350, 499]$	-1.672 ± 0.004
$n \in [400, 499]$	-1.671 ± 0.008
Fit range	Estimate for ζ
$n \in [50, 150]$	-1.6643 ± 0.0011
$n \in [100, 200]$	-1.6672 ± 0.0013
$n \in [150, 250]$	-1.6729 ± 0.0016
$n \in [200, 300]$	-1.6761 ± 0.0019
$n \in [250, 350]$	-1.6714 ± 0.0029
$n \in [300, 400]$	-1.670 ± 0.006
$n \in [350, 450]$	-1.672 ± 0.008
$n \in [400, 499]$	-1.671 ± 0.008

Remark. Define $t_{\ell_1,\ell_2,n-\ell_1-\ell_2}$ to be the number of (not necessarily unknotted) thetas with arm lengths ℓ_1, ℓ_2 and $n-\ell_1-\ell_2$, ordered according to length (assuming the three lengths have the same parity). After conversation with Stuart Whittington [64], we expect that a pattern theorem can be used to show that, for fixed ℓ_1 and ℓ_2 and as $n \to \infty$, there exist positive constants A, B such that

$$Anp_n \leq t_{\ell_1,\ell_2,n-\ell_1-\ell_2} \leq Bnp_n \tag{4.23}$$

for even n, with a similar result holding for odd n. This implies that the critical exponent for (all) thetas with the two shortest arms of fixed length is equal to $\alpha - 2$. This argument can be applied to both the square and cubic lattices.

4.2. Distribution of arm-lengths

We next look at the distribution of arm-lengths in polydisperse thetas. Let $\langle \ell_1 \rangle_n$ be the average number of edges in the shortest arm of thetas of total size n. We can similarly consider $\langle \ell_2 \rangle_n$ and $\langle \ell_1 + \ell_2 \rangle_n = n - \langle \ell_3 \rangle_n$. We will use $\langle \ell_{12} \rangle_n$ to denote this latter value.

For ℓ_1 we use data generated in (I). It is not unreasonable to expect that

$$\langle \ell_1 \rangle_n \sim Cn^\sigma \tag{4.24}$$

for some constants C, σ, where σ is (possibly) universal and depends only on dimension. In 2D we assume a correction term of the form

$$\langle \ell_1 \rangle_n \sim Cn^\sigma \left(1 + \frac{s}{n}\right) \qquad \Rightarrow \qquad \log\langle \ell_1 \rangle_n \sim \log C + \sigma \log n + \frac{s}{n} \tag{4.25}$$

for a constant s. For 3D we consider both $\frac{1}{n}$ and $\frac{1}{\sqrt{n}}$ as possible correction terms. As with the scaling of the number of thetas, for the square and cubic lattices it is possible that the constants C and s depend on the parity of n. We will thus make use of the median sequence $\langle \ell_1 \rangle_n^* = \frac{1}{2}\left(\langle \ell_1 \rangle_n + \langle \ell_1 \rangle_{n+1}\right)$.

For $\langle \ell_2 \rangle_n$ we similarly anticipate

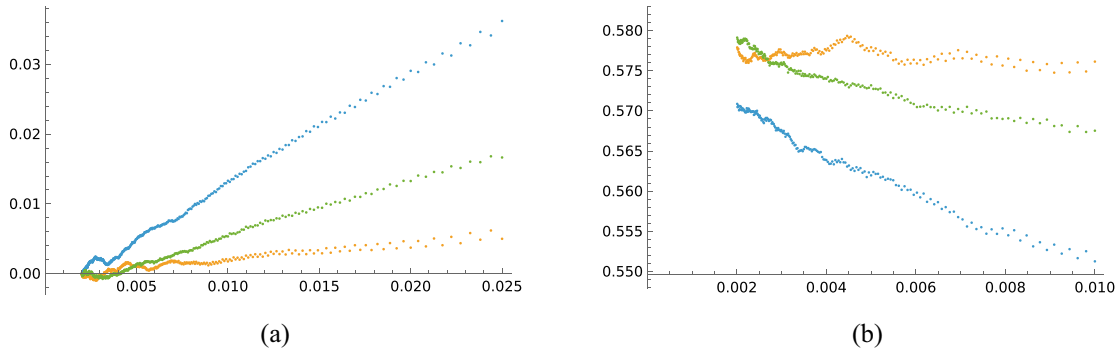
$$\langle \ell_2 \rangle_n \sim Dn^\tau \tag{4.26}$$

for constants D, τ. For numerical estimates we assume the same generic $\frac{1}{n}$ correction-to-scaling term. For $\langle \ell_{12} \rangle_n$, the exponent τ (if it exists) must be the same as for $\langle \ell_2 \rangle_n$, but the constant D may differ.

For the numerical analysis (estimating σ and τ) we performed essentially the same calculations as in section 4.1 when estimating ζ. We fit expressions of the form (4.25) or an expression similar to (4.17):

$$\frac{1}{\log 2}\left(\log\langle \ell_1 \rangle_n^* - \log\langle \ell_1 \rangle_{n/2}^*\right) \sim \sigma + \frac{s^\dagger}{n} \tag{4.27}$$

for a constant $s^\dagger$.



(a) (b)

Figure 12. (a) Plots of $\langle \ell_1 \rangle_n^* - 0.574 \log n$ (blue), $\langle \ell_2 \rangle_n^* - 0.579 \log n$ (orange) and $\langle \ell_{12} \rangle_n^* - 0.579 \log n$ (green) against $\frac{1}{n}$ for the square lattice. These have been shifted vertically so that the final term is 0. (b) Plots of the analogous quantities to (4.27), with the same colour schemes.

Table 4. Estimates of the exponents σ and τ for the square lattice, using fits of (4.27) (or analogous quantities for $\langle \ell_2 \rangle_n^*$ and $\langle \ell_{12} \rangle_n^*$) against $\frac{1}{n}$.

Fit range	Estimate for σ	Estimate for τ from $\langle \ell_2 \rangle_n^*$	Estimate for τ from $\langle \ell_{12} \rangle_n^*$
$n \in [50, 499]$	0.57348 ± 0.00034	0.57815 ± 0.00029	0.57969 ± 0.00032
$n \in [100, 499]$	0.5753 ± 0.0005	0.5773 ± 0.0004	0.5808 ± 0.0005
$n \in [150, 499]$	0.5760 ± 0.0007	0.5766 ± 0.0006	0.5819 ± 0.0007
$n \in [200, 499]$	0.5770 ± 0.0011	0.5751 ± 0.0009	0.5828 ± 0.0011
$n \in [250, 499]$	0.5779 ± 0.0017	0.5758 ± 0.0014	0.5843 ± 0.0016
$n \in [300, 499]$	0.5778 ± 0.0028	0.5762 ± 0.0022	0.5857 ± 0.0026
$n \in [350, 499]$	0.576 ± 0.005	0.578 ± 0.004	0.587 ± 0.004
$n \in [400, 499]$	0.574 ± 0.010	0.579 ± 0.007	0.585 ± 0.009

Square lattice

For the square lattice we use the same approach as in the previous section, this time to estimate σ and τ. The results are plotted in figure 12. By plotting $\langle \ell_1 \rangle_n^* - \sigma \log n$ against $\frac{1}{n}$ for different values of σ, and minimising the total residuals between these data and a line of best fit, we arrive at $\sigma = 0.574$. (See figure 12(a).) For τ we do the same with $\langle \ell_2 \rangle_n^*$ and $\langle \ell_{12} \rangle_n^*$ (we fit both separately using the same τ value, and added the residuals for both). This gave the best estimate $\tau = 0.579$. In figure 12(b) we plot the ratio quantity (4.27) and similarly for ℓ_2 and ℓ_{12}. We have fit these data against $\frac{1}{n}$, with the results presented in table 4.

From all these plots and the fits in table 4, we think it more likely than not that $\sigma_{\text{sq}} = \tau_{\text{sq}}$, with a value of 0.579 ± 0.015. We do note that $\frac{37}{64} \approx 0.578125$ is definitely in the vicinity of our estimate.

In figure 14(a) we plot the distribution of ℓ_1 (i.e. the fraction of size n thetas with a given value of ℓ_1) for a range of n. Note the vertical scale—the numbers drop off rapidly as ℓ_1 increases.

Cubic lattice

For the cubic lattice we repeat the calculations above. This time using the correction exponent $\Delta = \frac{1}{2}$ yields much straighter plots than $\Delta = 1$. See figure 13.

In figure 13(a) we get the best straight fit when $\sigma = 0.779$ and $\tau = 0.769$. Of course it is nonsensical for $\sigma > \tau$, but we expect that this anomaly just comes down to numerical uncertainty and a small-n effect. Figure 13(b) is analogous to figure 12(b), but plotted against $\frac{1}{\sqrt{n}}$.

In table 5 we present estimates for σ and τ on the cubic lattice, derived from fits of (4.27) (or analogous quantities for $\langle \ell_2 \rangle_n^*$ and $\langle \ell_{12} \rangle_n^*$) against $\frac{1}{\sqrt{n}}$. Note that if these exponents exist then we must have $\sigma \leq \tau$, however these estimates do not take into account any kind of constant amplitudes (e.g. C and D in (4.24) and (4.26)) which may vary dramatically between $\langle \ell_1 \rangle_n$, $\langle \ell_2 \rangle_n$ and $\langle \ell_{12} \rangle_n$.

Based on the estimates in table 5 (we are inclined to disregard the last row in favour of following the broad trend), our conservative estimate is that $\sigma_{\text{cu}} = \tau_{\text{cu}} = 0.769 \pm 0.022$.

In figure 14(b) we plot the distribution of ℓ_1 for thetas of various sizes on the cubic lattice. While the drop is not quite as precipitous as for the square lattice, cubic lattice thetas are still very much dominated by those with small ℓ_1.

We note here that our estimated range for σ and τ includes the estimated exponent $t = 0.75$ for the average size of the knotted component of a prime knot, as reported in [47]. It is however a fair

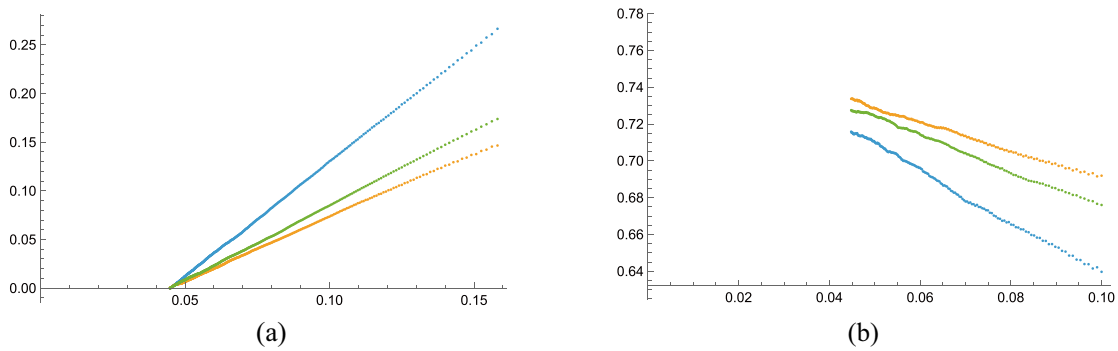


Figure 13. (a) Plots of $\langle \ell_1 \rangle_n^* - 0.779 \log n$ (blue), $\langle \ell_2 \rangle_n^* - 0.769 \log n$ (orange) and $\langle \ell_{12} \rangle_n^* - 0.769 \log n$ (green) against $\frac{1}{\sqrt{n}}$ for the cubic lattice. These have been shifted vertically so that the final term is 0. (b) Plots of the analogous quantities to (4.27), with the same colour schemes, plotted against $\frac{1}{\sqrt{n}}$. Linear fits have intercepts 0.781 (blue), 0.768 (orange) and 0.773 (green).

Table 5. Estimates of the exponents σ and τ for the cubic lattice, using fits of (4.27) (or analogous quantities for $\langle \ell_2 \rangle_n^*$ and $\langle \ell_{12} \rangle_n^*$) against $\frac{1}{\sqrt{n}}$.

Fit range	Estimate for σ	Estimate for τ from $\langle \ell_2 \rangle_n^*$	Estimate for τ from $\langle \ell_{12} \rangle_n^*$
$n \in [50, 499]$	0.7802 ± 0.0010	0.7654 ± 0.0009	0.7707 ± 0.0009
$n \in [100, 499]$	0.7814 ± 0.0013	0.7682 ± 0.0013	0.7726 ± 0.0012
$n \in [150, 499]$	0.7829 ± 0.0018	0.7689 ± 0.0018	0.7726 ± 0.0017
$n \in [200, 499]$	0.7822 ± 0.0026	0.7689 ± 0.0025	0.7707 ± 0.0024
$n \in [250, 499]$	0.778 ± 0.004	0.771 ± 0.004	0.7691 ± 0.0035
$n \in [300, 499]$	0.776 ± 0.006	0.774 ± 0.006	0.766 ± 0.006
$n \in [350, 499]$	0.771 ± 0.010	0.781 ± 0.010	0.757 ± 0.009
$n \in [400, 499]$	0.758 ± 0.021	0.785 ± 0.022	0.747 ± 0.020

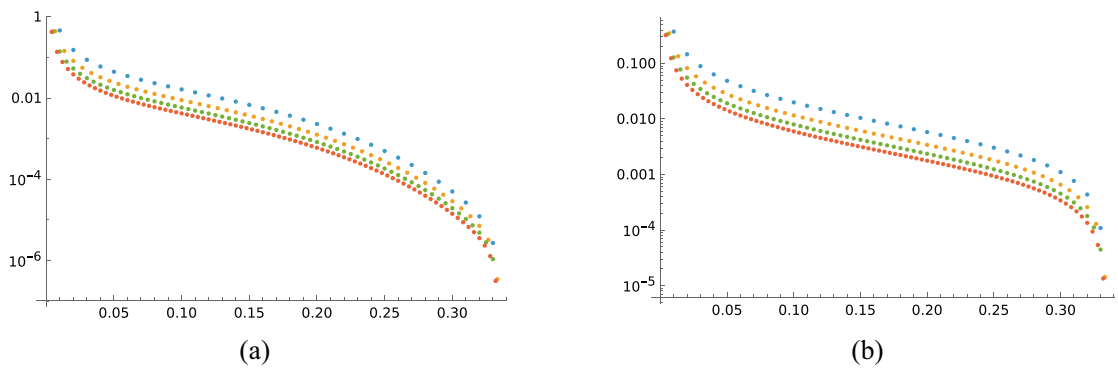


Figure 14. (a) The distribution of ℓ_1 (i.e. the fraction of thetas with a given ℓ_1) across square lattice thetas of size $n = 200$ (blue), 300 (orange), 400 (green) and 500 (red). The horizontal axis has been scaled by n. (b) The same plot for the cubic lattice.

bit larger than the other estimates (0.4 to 0.65) as outlined in section 1. The distribution of ℓ_1 in figure 14(b) can also be contrasted with the distribution of knot sizes in linear chains as found in [59], which showed a peak at knot-size around 100–300 (possibly independent of chain length), followed by a power-law decay. Here the distribution of ℓ_1 is monotone decreasing, with the smallest values being the most populous.

4.3. Separation of the branch points

Various geometric quantities for SAWs and polygons, such as the squared radius of gyration and squared end-to-end distance, are expected to scale (in mean) as a power law with exponent 2ν. For example, the squared radius of gyration scales as

$$\langle R_g^2 \rangle_n \sim C n^{2\nu} \left(1 + \frac{a}{n^{\Delta}}\right) \tag{4.28}$$

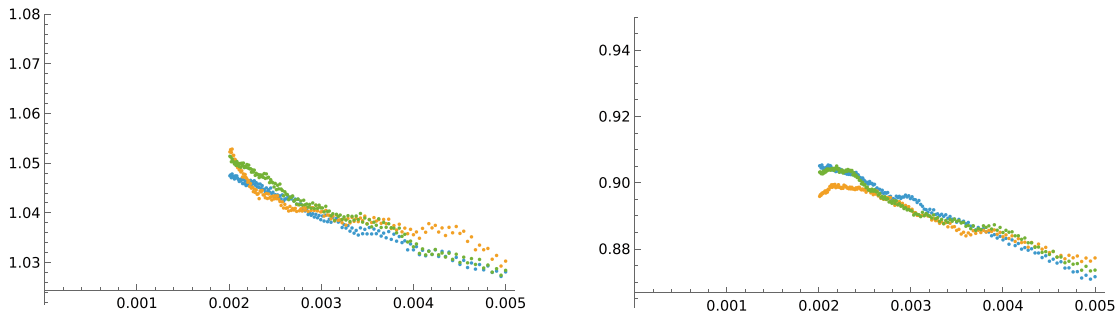


Figure 15. Ratio plots for estimating 2η on the (a) square lattice and (b) cubic lattice, both plotted against $\frac{1}{n}$ and using the median data in all cases. The Metropolis–Hastings transition probabilities were calculated using estimates from (I) (blue), (II) (orange) and (III) (green).

for constants C and a. The exponent ν is believed to be universal, taking values $\nu = \frac{3}{4}$ in two dimensions and $\nu = 0.587597(7)$ in three dimensions [8].

Here we investigate the separation of the two vertices of degree 3. For a theta T we write $D^2(T)$ to be the squared distance between the two vertices of degree 3. Then it is reasonable to expect that the mean of this quantity (across all thetas of size n) is

$$\langle D^2 \rangle_n \sim C n^{2\eta} \tag{4.29}$$

for some constants C and η. It is our goal here to estimate η.

To facilitate this calculation, we took the approximate theta counts generated in (I)–(III) and ran Markov chains with transition probabilities given by the Metropolis–Hastings algorithm (with Hastings factors as per (3.9)). For each of (I)–(III) we ran 20 independent Markov chains, collecting 10^9 samples in each, for the square and the cubic lattices. The target distribution was the uniform distribution across the respective state spaces: for example, for (I), the goal was to visit each state (n, ℓ_1) (approximately) equally often. We note that there is of course uncertainty in the approximate theta counts which we used to guide the Markov chains here, and expect that this will be reflected in further uncertainty in the estimates for η.

As in the previous sections there is dependence on the parity of n, so we also make use of the median sequence

$$\langle D^2 \rangle_n^* = \frac{1}{2}\left(\langle D^2 \rangle_n + \langle D^2 \rangle_{n+1}\right). \tag{4.30}$$

We used several methods for estimating 2η, including directly fitting to an expression of the form (4.29) as well as using the same kind of ratio method as in (4.27). See figure 15 for ratio plots.

In table 6 we give the estimates of 2η for the square and cubic lattices, fitting against $\frac{1}{n}$ for the square lattice and both $\frac{1}{n}$ and $\frac{1}{\sqrt{n}}$ for the cubic lattice.

Based on these data we make the conservative estimates $2\eta_{\text{sq}} = 1.072 \pm 0.010$ and $2\eta_{\text{cu}} = 0.94 \pm 0.02$.

We note that the number of edges between the two vertices of degree 3 is typically $O(n^\sigma)$, where estimates for σ were found in the previous section. Then

$$\frac{\eta}{\sigma} \approx \begin{cases} 0.927 \pm 0.033, & \text{square} \\ 0.61 \pm 0.03, & \text{cubic.} \end{cases} \tag{4.31}$$

These values should be contrasted with the values for ν. They suggest that in two dimensions, the vertices of degree 3 are, on average, further apart than would be 'typical' for two vertices separated by $O(n^\sigma)$ edges (say, in a SAW or a SAP). On the other hand, in three dimensions the value of $\frac{\eta}{\sigma}$ seems to be quite close to ν, suggesting that there is little or no additional repulsion between the vertices of degree 3.

Table 6. Estimates for 2η for square and cubic lattice thetas.

Fit range	Estimate for $2\eta_{\text{sq}}$
$n \in [50, 499]$	1.0595 ± 0.0004
$n \in [100, 499]$	1.0612 ± 0.0005
$n \in [150, 499]$	1.0623 ± 0.0007
$n \in [200, 499]$	1.0628 ± 0.0009
$n \in [250, 499]$	1.0648 ± 0.0013
$n \in [300, 499]$	1.0684 ± 0.0019
$n \in [350, 499]$	1.0721 ± 0.0032
$n \in [400, 499]$	1.075 ± 0.007

	Estimate for $2\eta_{\text{cu}}$	
Fit range	Using $\frac{1}{n}$	Using $\frac{1}{\sqrt{n}}$
$n \in [50, 499]$	0.9210 ± 0.0004	0.9643 ± 0.0007
$n \in [100, 499]$	0.9227 ± 0.0005	0.9611 ± 0.0009
$n \in [150, 499]$	0.9234 ± 0.0007	0.9577 ± 0.0013
$n \in [200, 499]$	0.9239 ± 0.0009	0.9550 ± 0.0017
$n \in [250, 499]$	0.9244 ± 0.0013	0.9535 ± 0.0025
$n \in [300, 499]$	0.9248 ± 0.0020	0.953 ± 0.004
$n \in [350, 499]$	0.9215 ± 0.0034	0.944 ± 0.007

5. Results: monodisperse theta-graphs

In this section we consider *monodisperse* theta-graphs, where the three arms have the same length. Since this requires the total length to be a multiple of 3, we will relax the definition slightly, and define two related sequences. First define

$$M_1(n) = \max_{T \in \mathcal{T}_n} \{\ell_1(T)\}, \tag{5.1}$$

$$M_{12}(n) = \max_{T \in \mathcal{T}_n} \{\ell_1(T) + \ell_2(T)\}. \tag{5.2}$$

It is not difficult to determine $M_1(n)$ and $M_{12}(n)$ for the square and cubic lattices. Then let

$$\overline{\theta}_n^{[1]} = |\{T \in \mathcal{T}_n : \ell_1(T) = M_1(n)\}| \tag{5.3}$$

$$\overline{\theta}_n^{[12]} = |\{T \in \mathcal{T}_n : \ell_1(T) + \ell_2(T) = M_{12}(n)\}|. \tag{5.4}$$

That is, $\overline{\theta}_n^{[1]}$ counts those theta-graphs of size n whose shortest arm is as long as possible, while $\overline{\theta}_n^{[12]}$ counts those for which the sum of the two shortest arms is as large as possible (equivalently, the longest arm is as short as possible). The relationship between these depend on whether n (mod 3) is equal to $0, 1, 2$. It is easy to show that

$$\overline{\theta}_n^{[1]} = \overline{\theta}_n^{[12]} \qquad \text{if } n \equiv 0 \pmod 3 \text{ and } n \geqslant 21 \text{ (square lattice) or } n \geqslant 9 \text{ (cubic lattice)}. \tag{5.5}$$

As an interesting aside, based on data from (I) and (II) we make the following conjecture, for which we have no combinatorial explanation. See figure 16.

Conjecture 1. For both the square and cubic lattices, as $n \to \infty$,

$$\overline{\theta}_n^{[12]} \sim \begin{cases} \frac{1}{2}\overline{\theta}_n^{[1]} & \text{if } n \equiv 1 \pmod 3 \\ 2\overline{\theta}_n^{[1]} & \text{if } n \equiv 2 \pmod 3. \end{cases} \tag{5.6}$$

For 'true' monodisperse thetas (i.e. those with all arms of exactly the same length, or equivalently those with total length $n \equiv 0 \pmod 3$), it is known that [43, 51]

$$\lim_{n \to \infty} \frac{1}{n} \log \overline{t}_n^{[1]} = \log \mu \tag{5.7}$$

$$\lim_{n \to \infty} \frac{1}{n} \log \overline{\theta}_n^{[1]} = \log \mu_0 \tag{5.8}$$

where both limits are taken through n divisible by 3.

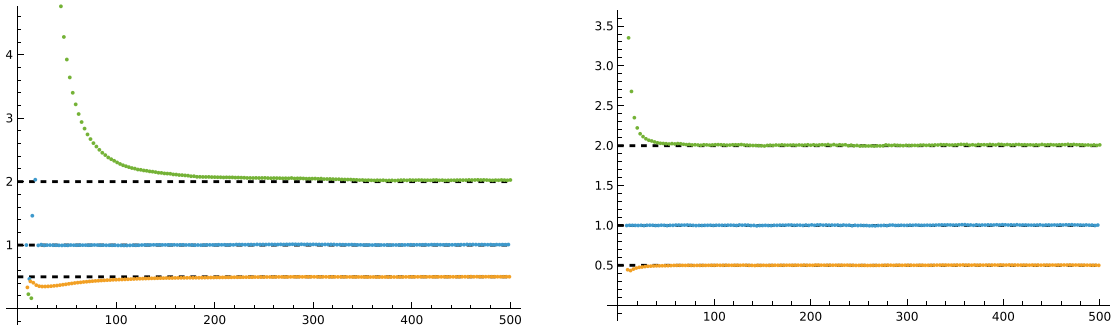


Figure 16. The ratio $\overline{\theta}_n^{[12]}/\overline{\theta}_n^{[1]}$ for the square lattice (a) and the cubic lattice (b), with $n \equiv 0 \pmod 3$ (blue), $n \equiv 1 \pmod 3$ (orange), and $n \equiv 2 \pmod 3$ (green). The ratios were computed by first averaging over the 20 independent Wang-Landau runs.

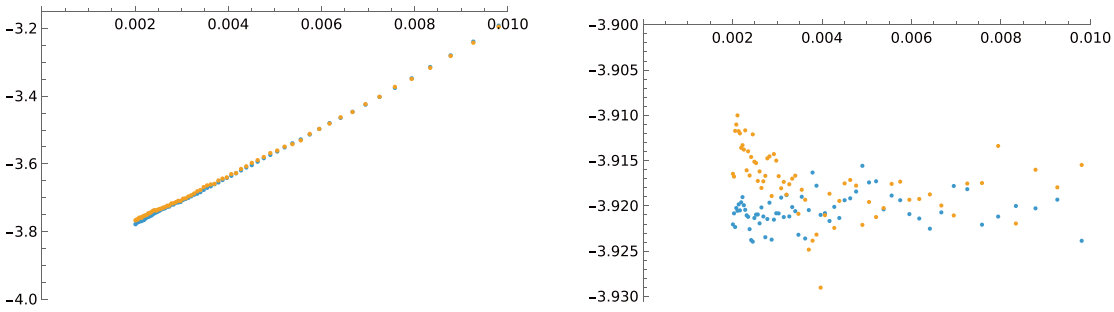


Figure 17. (a) A ratio plot for $\overline{\theta}_n^{[1]}$ (blue) and $\overline{\theta}_n^{[12]}$ (orange) on the square lattice, plotted against $\frac{1}{n}$ (using only values of n which are multiples of 6). (b) The equivalent plot for the cubic lattice.

For monodisperse thetas and for all values of $n \pmod 3$ we will assume asymptotics of the standard form

$$\overline{\theta}_n^{[1]} \sim C\mu_0^n n^\beta \tag{5.9}$$

for some exponent β, and where C may depend on the value of $n \pmod 3$. In light of conjecture 1, a similar expression should hold for $\overline{\theta}_n^{[12]}$. It is our goal in this section to estimate β.

Square lattice

Two-dimensional polymer networks with a fixed topology were studied by Duplantier in [18] using renormalisation theory and conformal invariance. Thetas are a particular case of 'watermelons' as considered there. For monodisperse thetas, the exponent β as in (5.9) corresponds to setting $L = 3$ in [18, Eqn. 23]:

$$\frac{20 - 9L^2}{32} - (L-1) = -\frac{125}{32} \approx -3.90625. \tag{5.10}$$

We have performed similar analyses as in the previous sections for estimating β. Plotting $\overline{\theta}_n^{[1]} - n\log\mu_0 - \beta\log n$ against $\frac{1}{n}$ for different values of β (note that because we are only working with values of n which are multiples of 3, we do not use the median data here), and likewise for $\overline{\theta}_n^{[12]}$, and then finding the value which gives the most linear data (plot omitted for brevity) gives $\beta = -3.912$ as the optimal value. Using the ratio technique (see figure 17(a)) gives plots which are not quite linear for small n, but trend towards linearity for large n. We present estimates for β using fits against $\frac{1}{n}$ in table 7.

Based on the estimates in table 7, we estimate $\beta_{\text{sq}} = -3.901 \pm 0.010$, strongly supporting the prediction of Duplantier [18] that $\beta_{\text{sq}} = -\frac{125}{32}$.

Cubic lattice

For the cubic lattice we attempted the same analysis, see figure 17(b). We are unable to determine whether $\Delta = \frac{1}{2}$ or $\Delta = 1$ is the more appropriate correction-to-scaling exponent here. Estimates for β_{cu} are presented in table 7, using fits against both $\frac{1}{n}$ and $\frac{1}{\sqrt{n}}$. To be conservative we take the last estimate $\beta_{\text{cu}} = -3.905 \pm 0.019$. It is intriguingly not impossible that $\beta_{\text{cu}} = \beta_{\text{sq}}$ and so independent of dimension.

Table 7. Estimates for β for square and cubic lattice thetas.

Fit range	Estimate for β_{sq}
$n \in [48, 498]$	-3.9821 ± 0.0028
$n \in [96, 498]$	-3.9237 ± 0.0015
$n \in [144, 498]$	-3.9132 ± 0.0020
$n \in [192, 498]$	-3.9094 ± 0.0028
$n \in [240, 498]$	-3.904 ± 0.004
$n \in [288, 498]$	-3.900 ± 0.006
$n \in [336, 498]$	-3.901 ± 0.009

	Estimate for β_{cu}	
Fit range	Using $\frac{1}{n}$	Using $\frac{1}{\sqrt{n}}$
$n \in [48, 498]$	-3.9195 ± 0.0009	-3.9200 ± 0.0017
$n \in [96, 498]$	-3.9182 ± 0.0013	-3.9170 ± 0.0026
$n \in [144, 498]$	-3.9169 ± 0.0019	-3.914 ± 0.004
$n \in [192, 498]$	-3.9157 ± 0.0027	-3.912 ± 0.005
$n \in [240, 498]$	-3.912 ± 0.004	-3.906 ± 0.008
$n \in [288, 498]$	-3.913 ± 0.006	-3.907 ± 0.011
$n \in [336, 498]$	-3.912 ± 0.009	-3.905 ± 0.019

6. Conclusion and outlook

In this work we have studied polydisperse two-dimensional (square lattice) and unknotted three-dimensional (simple cubic lattice) lattice embeddings of theta graphs using a combination of the Wang-Landau Monte Carlo method and local BFACF moves to randomly sample configurations. Our results point to non-trivial values of the entropic exponents, and new exponents for the arm lengths and the separation of the vertices of degree three (branch points). We estimate the entropic exponent for the square lattice to be $\zeta_{sq} = -1.470 \pm 0.011$ whilst on the cubic lattice we find $\zeta_{cu} = -1.67 \pm 0.015$. These are distinct from the values $\alpha - 2$ (-1.5 and -1.763 respectively), which would result from taking SAPs of length n and inserting a small loop in one of n possible positions.

For the 'short' arm and sum of two shortest arms length exponent we find there is likely only one value and that is 0.579 ± 0.015 on the square lattice and 0.769 ± 0.022 on the cubic lattice. For the mean squared distance between the branch points, we estimate exponents of 0.536 ± 0.005 on the square lattice and 0.47 ± 0.01 for the cubic lattice.

We have also looked at the entropic exponents in the monodisperse cases and our results support the previous prediction of [18, 19] in two dimensions. We note that our estimate for this exponent on cubic lattice is very close to the square lattice result and exact prediction: this calls for further examination. Considering future directions it would be interesting to know whether the exponents in the polydisperse case in two dimensions have exact values that can be predicted. Further work will need to be undertaken to understand the relationship of our result to those of knotted polygons.

While it is in principle possible to incorporate non-local moves like pivots into a Monte Carlo method for thetas, the vertices of degree 3 would make these quite challenging to implement. Even with only local BFACF-type moves, there are two fairly straightforward potential extensions of this work:

- Other objects: tadpoles, dumbbells, 3-stars and other spatial graphs with vertices of degree 3 can easily be sampled using the same methodology.
- Other lattices: We expect that a similar set of moves to those of figure 5 involving vertices of degree 3 can be implemented for the triangular and FCC lattices (in fact they are simpler).

Finally, to reiterate here in three dimensions we studied *unknotted* theta-graphs. There have now been studies of how knots affect SAPs and in this work how the theta topology modifies the scaling behaviour. It is not entirely clear whether different knot types of thetas would behave in the same manner as unknotted thetas, and so the interaction of knottedness with the theta topology would be of real interest.

Acknowledgments

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Data availability statement

The data that support the findings of this study will be openly available following an embargo at the following URL/DOI: www.nicholasbeaton.com [67].

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References

[1] Atisattapong W and Marupanthorn P 2021 Wang–Landau sampling for estimation of the reliability of physical networks *Comput. Phys. Commun.* **262** 107831
[2] Baiesi M, Orlandini E and Stella A L 2010 The entropic cost to tie a knot *J. Stat. Mech.* 06012
[3] Beaton N R, Ishihara K, Atapour M, Eng J W, Vazquez M, Shimokawa K and Soteros C E 2026 Entanglement statistics of polymers in a lattice tube and unknotting of 4-plats *Discrete Appl. Math.* **379** 242–71
[4] Belardinelli R E and Pereyra V D 2007 Fast algorithm to calculate density of states *Phys. Rev.* E **75** 046701
[5] Berg B and Foerster D 1981 Random paths and random surfaces on a digital computer *Phys. Lett.* B **106** 323–6
[6] Bradly C J, Van Rensburg E J J, Owczarek A L and Whittington S G 2019 Force-induced desorption of 3-star polymers in two dimensions *J. Phys. A: Math. Theor.* **52** 315002
[7] Caracciolo S, Guttmann A J, Jensen I, Pelissetto A, Rogers A N and Sokal A D 2005 Correction-to-scaling exponents for two-dimensional self-avoiding walks *J. Stat. Phys.* **120** 1037–100
[8] Clisby N 2010 Accurate estimate of the critical exponent ν for self-avoiding walks via a fast implementation of the pivot algorithm *Phys. Rev. Lett.* **104** 055702
[9] Clisby N 2013 Calculation of the connective constant for self-avoiding walks via the pivot algorithm *J. Phys. A: Math. Theor.* **46** 245001
[10] Clisby N and Dünweg B 2016 High-precision estimate of the hydrodynamic radius for self-avoiding walks *Phys. Rev.* E **94** 052102
[11] Clisby N and Jensen I 2012 A new transfer-matrix algorithm for exact enumerations: self-avoiding polygons on the square lattice *J. Phys. A: Math. Theor.* **45** 115202
[12] Dabrowski-Tumanski P, Goundaroulis D, Stasiak A, Rawdon E J and Sulkowska J I 2024 Theta-curves in proteins *Protein Sci.* **33** e5133
[13] de Carvalho C A ao and Caracciolo S 1983 A new Monte-Carlo approach to the critical properties of self-avoiding random walks *J. Physique* **44** 323–31
[14] de Carvalho C A ao, Caracciolo S and Fröhlich J 1983 Polymers and $g|\varphi|^4$ theory in four dimensions *Nucl. Phys.* B **215** 209–48
[15] De Gennes P 1979 *Scaling Concepts in Polymer Physics* (Cornell University Press)
[16] Delbruck M and Fuller F B 1962 Mathematical problems in the biological sciences *Proc. Symp. on Applied Mathematics* vol 14 (American Mathematical Society) pp 55–63
[17] Diao Y 1993 Minimal knotted polygons on the cubic lattice *J. Knot Theory Ramifications* **02** 413–25
[18] Duplantier B 1986 Polymer network of fixed topology: renormalization, exact critical exponent γ in two dimensions and $d = 4 - \varepsilon$ *Phys. Rev. Lett.* **57** 2332
[19] Duplantier B 1989 Statistical mechanics of polymer networks of any topology *J. Stat. Phys.* **54** 581–680
[20] Farago O, Kantor Y and Kardar M 2002 Pulling knotted polymers *Europhys Lett.* **60** 53
[21] Frisch H L and Wasserman E 1961 Chemical topology *J. Am. Chem. Soc.* **83** 3789–95
[22] Gleeson J 2024 Monte Carlo enumeration of topological polymers *Master's Thesis* University of Melbourne
[23] Gu Y, Zhao J and Johnson J A 2020 Polymer networks: from plastics and gels to porous frameworks *Angew. Chem., Int. Edn* **59** 5022–49
[24] Guttmann A J 2009 *Polygons, Polyominoes and Polycubes* (*Lecture Notes in Physics*) (Springer)
[25] Guttmann A J and Sykes M F 1973 Limiting ring closure probability index for the self avoiding random walk problem *J. Phys. C: Solid State Phys.* **6** 945
[26] Guttmann A J and Whittington S G 1978 Two-dimensional lattice embeddings of connected graphs of cyclomatic index two *J. Phys. A: Math. Gen.* **11** 721–9
[27] Guttmann A J 2021 private communication
[28] Guttmann A J, Parviainen R and Rechnitzer A 2004 Self-avoiding walks and trails on the 3.12^2 lattice *J. Phys. A: Math. Gen.* **38** 543

[29] Jacobsen J L, Scullard C R and Guttmann A J 2016 On the growth constant for square-lattice self-avoiding walks *J. Phys. A: Math. Theor.* **49** 494004
[30] Janse van Rensburg E J 2008 Thoughts on lattice knot statistics *J. Math. Chem.* **45** 7
[31] Janse van Rensburg E J 2009 Monte Carlo methods for the self-avoiding walk *J. Phys. A: Math. Theor.* **42** 323001
[32] Janse van Rensburg E J 2015 *The Statistical Mechanics of Interacting Walks, Polygons, Animals and Vesicles (Oxford Lecture Series in Mathematics and Its Applications)* 2nd edn (Oxford University Press)
[33] Janse van Rensburg E J and Rechnitzer A 2011 BFACF-style algorithms for polygons in the body-centered and face-centered cubic lattices *J. Phys. A: Math. Theor.* **44** 165001
[34] Janse van Rensburg E J and Rechnitzer A 2011 On the universality of knot probability ratios *J. Phys. A: Math. Theor.* **44** 162002
[35] Janse van Rensburg E J and Rechnitzer A 2012 Generalized atmospheric sampling of knotted polygons *J. Knot Theory Ramifications* **20** 1145–71
[36] Janse van Rensburg E J and Whittington S G 1990 The knot probability in lattice polygons *J. Phys. A: Math. Gen.* **23** 3573
[37] Janse van Rensburg E J and Whittington S G 1991 The BFACF algorithm and knotted polygons *J. Phys. A: Math. Gen.* **24** 5553–67
[38] Janse van Rensburg E J and Whittington S G 1991 The dimensions of knotted polygons *J. Phys. A: Math. Gen.* **24** 3935
[39] Janse van Rensburg E J and Whittington S G 2024 Exponential growth rate of lattice comb polymers *J. Phys. A: Math. Theor.* **57** 485002
[40] Kim H, Lee H J, No S, Oh S and Yoo H 2022 Lattice conformation of theta-curves accompanied with Brunnian property *J. Phys. A: Math. Theor.* **55** 435207
[41] Landau D P, Tsai S-H and Exler M 2004 A new approach to Monte Carlo simulations in statistical physics: Wang-Landau sampling *Am. J. Phys.* **72** 1294–302
[42] Li Y, Wüst T, Landau D and Lin H 2007 Numerical integration using Wang–Landau sampling *Comput. Phys. Commun.* **177** 524–9
[43] Madras N, Sumners D W and Whittington S G 2009 Almost unknotted embeddings of graphs in Z^3 and higher dimensional analogues *J. Knot Theory Ramif.* **18** 1031–48
[44] Madras N and Slade G 1996 *The Self-Avoiding Walk* (Birkhäuser)
[45] Malakis A, Martinos S S, Hadjiagapiou I A and Peratzakis A S 2004 On the Wang–Landau method using the N-fold way *Int. J. Mod. Phys.* C **15** 729–40
[46] Mansfield M L and Douglas J F 2010 Properties of knotted ring polymers: I. Equilibrium dimensions *J. Chem. Phys.* **133** 044903
[47] Marcone B, Orlandini E, Stella A L and Zonta F 2007 Size of knots in ring polymers *Phys. Rev.* E **75** 041105
[48] Moriuchi H 2009 An enumeration of theta-curves with up to seven crossings *J. Knot Theory Ramif.* **18** 167–97
[49] No S, Oh S and Yoo H 2021 Topological aspects of theta-curves in cubic lattice *J. Phys. A: Math. Theor.* **54** 455204
[50] Orlandini E, Stella A L and Vanderzande C 2009 The size of knots in polymers *Phys. Biol.* **6** 025012
[51] Soteros C E 1992 Adsorption of uniform lattice animals with specified topology *J. Phys. A: Math. Gen.* **25** 3153
[52] Soteros C E, Sumners D W and Whittington S G 1992 Entanglement complexity of graphs in $\mathbb{Z}^3$ *Math. Proc. Cam. Phil. Soc.* **111** 75–91
[53] Sumners D W and Whittington S G 1988 Knots in self-avoiding walks *J. Phys. A: Math. Gen.* **21** 1689
[54] Sykes M F 1961 Some counting theorems in the theory of the Ising model and the excluded volume problem *J. Math. Phys.* **2** 52–62
[55] Tamaki K 2018 Knots and spatial graphs in the simple cubic lattice *Master's Thesis* Saitama University
[56] Taylor M P, Paul W and Binder K 2009 Phase transitions of a single polymer chain: A Wang–Landau simulation study *J. Chem. Phys.* **131** 114907
[57] Tezuka Y, Tsuchitani A, Yoshioka Y and Oike H 2003 Synthesis of θ-shaped poly(THF) by electrostatic self-assembly and covalent fixation with three-armed star telechelics having cyclic ammonium salt groups *Macromolecules* **36** 65–70
[58] Tubiana L *et al* 2024 Topology in soft and biological matter *Phys. Rep.* **1075** 1–137
[59] Tubiana L, Rosa A, Fragiacomo F and Micheletti C 2013 Spontaneous knotting and unknotting of flexible linear polymers: Equilibrium and kinetic aspects *Macromolecules* **46** 3669–78
[60] Uehara E and Deguchi T 2018 Statistical properties of multi-theta polymer chains *J. Phys. A: Math. Theor.* **51** 134001
[61] Virnau P, Kantor Y and Kardar M 2005 Knots in globule and coil phases of a model polyethylene *J. Am. Chem. Soc.* **127** 15102–6
[62] Wüst T and Landau D P 2009 Versatile approach to access the low temperature thermodynamics of lattice polymers and proteins *Phys. Rev. Lett.* **102** 178101
[63] Wang F and Landau D P 2001 Efficient, multiple-range random walk algorithm to calculate the density of states *Phys. Rev. Lett.* **86** 2050–3
[64] Whittington S G 2026 Private communication
[65] Yin J and Landau D 2012 Massively parallel Wang–Landau sampling on multiple GPUs *Comput. Phys. Commun.* **183** 1568–73
[66] Zhan L 2008 A parallel implementation of the Wang–Landau algorithm *Comput. Phys. Commun.* **179** 339–44
[67] Beaton N R 2026 Personal webpage (available at: www.nicholasbeaton.com)